



Statistics of detonation confinement: 1D, 2D and 3D simulations in hydrogen–oxygen

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ABSTRACT

Statistical analysis of one-dimensional (1D), two-dimensional (2D), and three-dimensional (3D) detonation simulations is performed to understand the role of confinement on detonation frontal velocity and thermodynamic states. Simulations considered include a 2D wide channel (5 detonation cells across), a 2D narrow channel (1.5 cells across), a 3D square channel (10 cells around perimeter), and a 3D round tube (6 cells around perimeter), with identical mixture conditions of stoichiometric hydrogen–oxygen with 3000 parts per million by volume (PPMv) ozone at 300 K and 15 kPa. 1D and 2D simulations with the same mixture without ozone are also considered. All simulations solve the compressible Navier–Stokes equations with detailed chemistry and exhibit good agreement with macroscopic characteristics of experiments (e.g., cell size). Results show that the frontal velocity in 3D simulations is more irregular as compared to the 2D simulations, which are more irregular compared to the 1D simulations. The irregularity in frontal velocity leads to broader distributions of shock velocities and shock decay behavior. The 3D simulations also exhibit a wider and more extreme set of thermodynamic states as compared to the 2D simulations immediately behind the shock wave, within the induction zone, and throughout the rest of the domain. Increasing relative boundary losses leads to quantitative differences in both thermodynamic state and frontal velocity, consistent with 1D steady models of detonation with losses. The conclusions drawn in this work are not sensitive to numerical grid resolution.

1. Introduction

Detonation chemistry and detonation confinement both play important roles on propagation, limits, and cellular dynamics. For example, while detonation mixture conditions (e.g., chemical composition, initial temperature, and initial pressure) are thought to generally control detonation cell size [1], substantial variations of observed cell size and appearance exist in the same mixture with varying confining geometry [2,3]. One tool past researchers have used to interpret detonation sensitivity to mixture chemistry is statistical analysis of wave velocity [4–6] due, in part, to its utility in analyzing conventional turbulent flows. Both Frederick and Shepherd [5,6] observed substantial differences in frontal velocity statistics across varying mixture chemistry. Mixtures that are less stable (i.e., higher reduced activation energy ϵ_i and/or lower ratio of specific heat γ_{VN} [7]) were shown to feature broader distributions of velocity. These previous observations, however, were limited to two-dimensional (2D) simulation or

quasi-2D channel experiments, and so, to our knowledge, little is known about how velocity statistics might vary with confining geometry. Although substantial work has been done to understand 3D detonations phenomenologically, including both detailed experiments and simulations (e.g., [8–12]), a detailed characterization of velocity is missing. Particularly, how velocity behavior may vary between 2D and three-dimensional (3D) domains is of interest for better understanding the nature of 3D detonation.

Detonation frontal velocity controls shock strength, and therefore the thermodynamic state immediately behind the shock wave. Several key detonation physics are sensitive to thermodynamic state such as ignition kinetics, real gas effects [13,14], and vibrational relaxation [15,16]. Quantifying the thermodynamic state distributions can inform modelers of the range of states expected in a simulation and help predict the importance of considering various physics. The thermodynamic state within the domain is also important for detonation applications. For example, peak pressures within detonating flows may

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constrain structural requirements, while peak temperatures influence heat transfer and material requirements [17].

This work aims to (1) isolate the role of confining geometry on the statistical distribution of detonation frontal velocity, and (2) understand how this variability manifests in thermodynamic state, critical for detonation applications. To determine the role of confinement on detonation statistics, this work relies on well-resolved 1D, 2D, and 3D detonation simulations. We consider simulations solving the compressible Navier–Stokes equations with detailed chemistry and realistic boundary conditions (no-slip and isothermal walls). Two mixtures are considered, stoichiometric hydrogen–oxygen at 15 kPa initial pressure and 300 K initial temperature, and the same mixture with 3000 parts per million by volume (PPMv) of ozone (O_3) added. Ozone is shown to reduce detonation cell size [18] and stabilize detonation by reducing its effective activation energy [7]. These mixtures are chosen because they are well studied experimentally and kinetically. Some of these simulations are described in our previous work [2], where detailed structural characterization is presented. We use this structural characterization to interpret the statistical results presented herein.

2. Numerical solver and setup

The simulations are performed with the in-house code *PsiPhi* [19, 20] that solves the conservation equations for mass, momentum, total internal energy, and partial densities (fully compressible multi component Navier–Stokes equations) to simulate detonation wave propagation. The equations are discretized on an equidistant Cartesian grid utilizing the Finite Volume Method. An approximate Riemann solver computes convective fluxes at cell interfaces with the aid of a monotonicity preserving reconstruction scheme featuring a theoretical accuracy of 5th order, while 2nd order central difference schemes are used for diffusive fluxes. A Strang Operator Splitting framework allows explicit time integration of convection and diffusion with a 3rd order accurate Runge–Kutta scheme, while the implicit solver CVODE advances the solution in time from reaction. Sub-filter dynamics are modeled with an eddy viscosity/diffusivity approach, and the filtered chemical source term is modeled with a tophat PDF for temperature to account for non-linear reaction rate constants. More details can be found in [2,19,20].

The reaction kinetics are modeled with the Foundational Fuel Chemistry Model Version 1.0 (FFCM-1) [21,22] with the Princeton ozone sub-model [23]. Two 3D geometries are considered: a square channel $12 \times 12 \times 12$ mm (3D square), and a tube 12 mm in diameter and 12 mm in length (3D tube); two 2D geometries: 24 mm in width and 36 mm in length (2D wide), and 6 mm in width and 36 mm in length (2D narrow); and one 1D geometry: 24 mm in length (1D). The 3D tube simulation is set up in a rectangular domain, where the effects of walls are modeled with the immersed boundary technique. The boundary conditions used are no-slip and isothermal (300 K) for the walls, a Dirichlet boundary condition for the inlet, and a partially reflecting boundary condition for the outlet. This setup allows the numerical domain to be moved at the average wave speed relative to the laboratory reference without wave reflection. The numerical resolution considered in all simulations is $10 \mu\text{m}$ (24 grid points per half reaction length for the ozonated mixture, and approximately 48 for the non-ozonated mixture) unless otherwise noted. The $10 \mu\text{m}$ 3D simulations are run on 64,000 cores for approximately 150 wall-clock hours, for a total cost of 10 million CPU-hours per simulation. The base mixture considered is ozonated hydrogen–oxygen, and a second mixture is considered, which is identical, except that it does not contain any ozone, for the 2D wide and 1D geometries. Note that a time-resolved 1D simulation of the base mixture ($2H_2-O_2$ with 3000 PPMv O_3) results in stable propagation, quantitatively similar to the Zel'dovich–von Neumann–Döring (ZND) solution. The non-ozonated mixture, however, is unstable in 1D, and so it is used for statistical analysis.

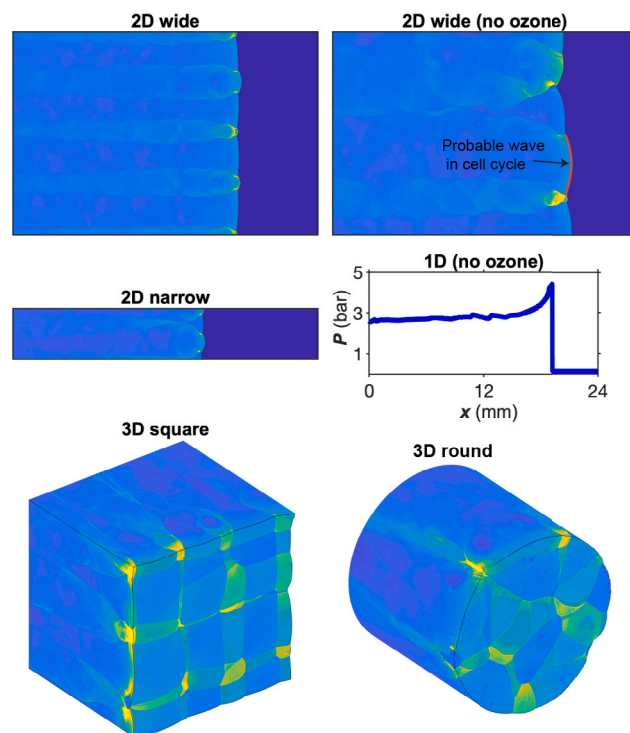


Fig. 1. Pressure pseudocolors of a single snapshot in time of all simulations considered in this work. The colorbar applies to all figures except the 1D case. The grid resolution in the 1D and 2D figures is $10 \mu\text{m}$, while in the 3D figures is $20 \mu\text{m}$.

3. Results and discussion

Fig. 1 presents the simulations considered in this work, showing pressure pseudocolor for 2D and 3D simulations, and a pressure trace for the 1D simulation. All snapshots are chosen from the well-converged portions of the simulations to exclude memory effects from initial conditions. Comparing the 2D wide simulation to the 3D square simulation, the basic Mach stem and incident shock structure is observed in both cases, but the 3D square simulation exhibits a more complex shock lattice structure due to the existence of a third dimension. As described in [2], the 3D tube simulation exhibits what appears to be a more chaotic frontal shock structure compared to the organized square lattice of the 3D square simulation. The 2D wide (no ozone) simulation exhibits a larger cell size, evident from the smaller number of triple points in the domain (both channel heights are 24 mm). The 2D narrow case appears similar in nature to the 2D wide case, while the 1D (no ozone) case presents a simple 1D detonation profile similar to the ZND solution.

Numerical grid convergence was considered extensively in past work [2], where key conclusions were shown to be insensitive to grid resolution. The highest resolution 2D and 3D simulations also closely match key macroscopic observable quantities from experiment (cell size and velocity) [2], giving further confidence in the veracity of the simulations. Further confirmation that the following statistical analyses below do not depend on numerical grid convergence is presented in Section 3.4.

3.1. Frontal velocity statistics

Frontal velocity statistics are determined by calculating the shock-normal velocity for the frontal shock in each time step of the simulation for each y location for 2D simulations and each y, z location for 3D simulations. The detailed methodology is described in Section S1 of the Supplementary Material. Example shock-normal frontal velocity,

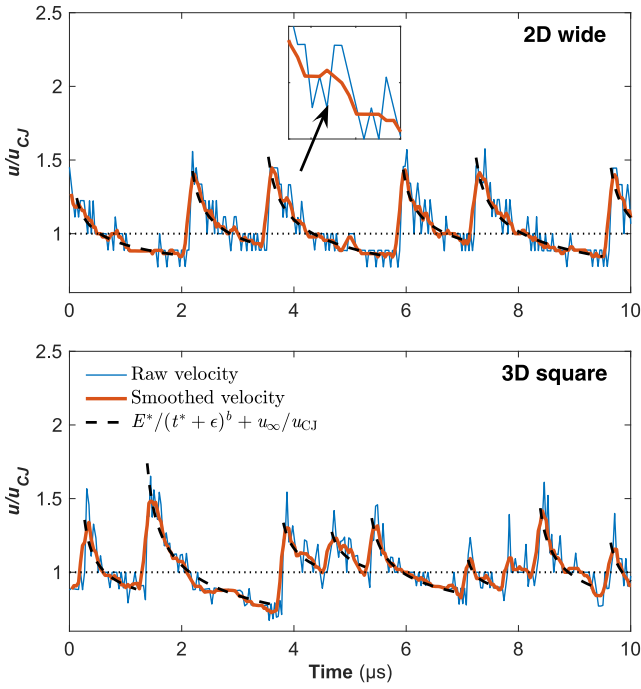


Fig. 2. Raw velocity, smoothed velocity, and fitted velocity for a line in the x -direction for the 2D wide case (top) and 3D square case (bottom). Inset on the top graph shows $t = 4.95$ to $5.65 \mu\text{s}$ and $u/u_{\text{CJ}} = 0.95$ to 1.3 .

Table 1
Normalized average propagation velocity.

Simulation	$\bar{u}/u_{\text{CJ}}$
2D wide	0.996
2D wide (no ozone)	1.000
2D narrow	0.974
1D (no ozone)	1.004
3D square	0.973
3D tube	0.971

normalized by the CJ velocity u_{CJ} , from a single y location for the 2D wide case and from a single y, z location for the 3D square case is presented in Fig. 2 as thin blue lines. Due to numerical discretization, and exacerbated because the numerical domain moves with the average wave speed, the velocity can only contain discrete values. To account for the expected smoother real behavior, the raw calculated velocity is smoothed with a moving mean filter, with a window size of 10 time steps ($dt \approx 0.036 \mu\text{s}$). The resultant smoothed normalized velocity is shown with thicker red lines in Fig. 2, and exhibits a smoothly varying velocity in time.

The example velocity traces suggest that the 2D wide case exhibits a regular velocity decay behavior, with consistent peak velocities ($u_0/u_{\text{CJ}} \approx 1.5$). The 3D square channel velocity, meanwhile, appears substantially more chaotic, with several small velocity decay profiles and one larger one. The differences between the geometries is presumably due to substantially more complex transverse waves in the 3D case, which interact in two directions ($y-z$), versus one direction (y) in 2D. To make more meaningful conclusions, all smoothed velocity values in the domain are considered statistically. Table 1 shows the average propagation velocity $\bar{u}$ for propagation within the well-converged duration of the simulation. Both 2D wide simulations and the 1D simulation propagate at nearly exactly the CJ velocity, while the 2D narrow, 3D square, and 3D tube all propagate at approximately 97% of the CJ velocity.

The kernel density estimation technique is used to estimate the probability distribution $P_i(u)$ of frontal velocity [6,24], corrected to

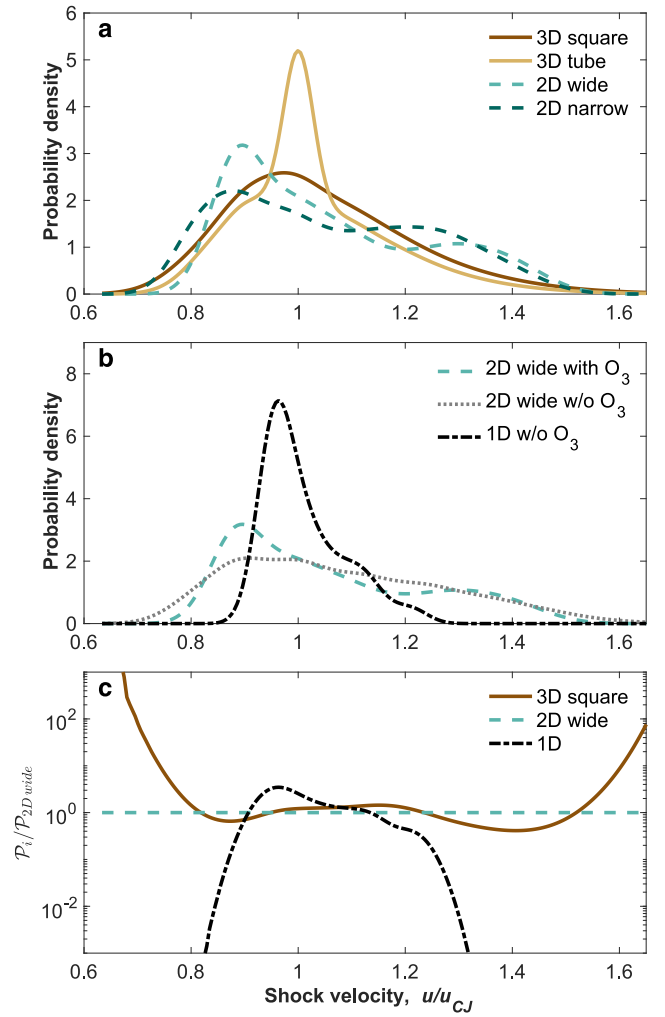


Fig. 3. Probability distribution of frontal velocity, for various confining geometries for the ozonated mixture a, for 2D wide with ozone, 2D wide without ozone, 1D without ozone b, and probability densities for 1D, 2D, and 3D simulations, normalized to 2D wide probability density in their identical mixture c.

be spatially-weighted. All calculated frontal velocity values within the domain are used, which constitutes $O(10^7)$ points in all cases, except for in the 2D narrow case and 1D case, where $O(10^6)$ and $O(10^3)$ measurements are available, respectively. The frontal velocity statistics are compared for identical mixtures with varying confining geometries in Fig. 3a. Several notable differences exist among the different confining geometries. First, the 2D wide and 2D narrow distributions appear similar, but the 2D narrow simulation is shifted to slower velocities. This is at least partially explained by the slower bulk velocities (see Table 1) from the increased relative role of boundary losses. Both 2D distributions exhibit a mild bimodal behavior, with a primary peak at about $0.85u_{\text{CJ}}$, and a smaller secondary peak at $1.3u_{\text{CJ}}$. The primary peak corresponds to the wave velocity within the cellular cycle where the wave is the widest (approximately where neighboring cells are being initiated), shown in red in Fig. 1 as the probable wave in the cellular cycle. The secondary peak is likely due to the frontal acceleration from the energy release of the kernel (where the initial acceleration is from strong jetting phenomena, observable in mixtures with relatively high ratio of specific heat, $\gamma = 1.32$ in this work) [25,26]. The secondary peak in velocity is seen in the 2D wide example velocity profile, magnified in the inset, where a small bump in velocity is observed. Comparing the 3D square to the 2D simulations, the overall shape is similar, without a secondary peak,

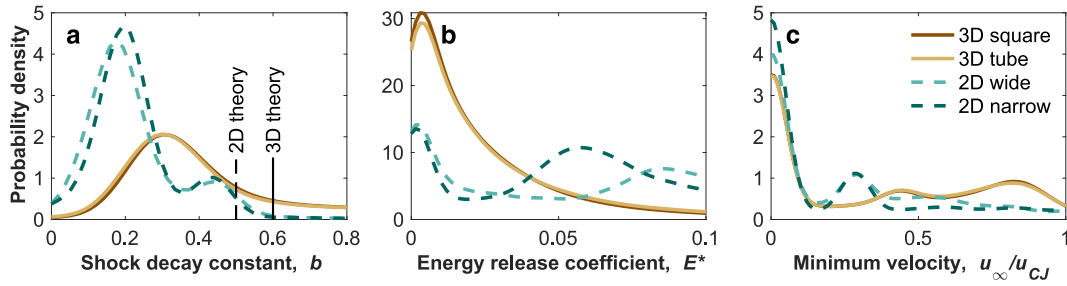


Fig. 4. Probability distribution of non-dimensionalized fitted power law decay function coefficients for Eq. (2), plotted for various confining geometries. The legend in c applies to all sub-figures.

and with a somewhat broader distribution, including larger tails at lower ($< 0.7u/u_{CJ}$) and higher ($> 1.5u/u_{CJ}$) velocity. The high velocity events are presumably due to the multi-kernels described in [2], not present in 2D simulations, while the low velocity points are presumably due to decreased confinement leading to faster velocity decay. The 3D tube simulation has a substantially different statistical appearance, with $u/u_{CJ} = 1$ much more frequently observed than in other simulations, and with smaller tails as compared to the 3D square simulation. This may be due to the more complex and chaotic shock behavior, leading to a more continuous velocity behavior.

Chemistry also affects the observed velocity behavior. Fig. 3b shows two simulations in identical geometries (2D wide), one with ozone and one without ozone. Quantified by reduced activation energy ϵ_i and ratio of specific heats at the von Neumann state γ_{VN} , the mixture with ozone ($\epsilon_i = 2.52$, $\gamma_{VN} = 1.32$) is more stable as compared to the mixture without ozone ($\epsilon_i = 4.41$, $\gamma_{VN} = 1.32$) [7]. Note that other stability criteria (e.g., the ratio of induction to exothermic length [27]) also predict the ozonated mixture to be more stable than the non-ozonated mixture. Consistent with what is reported previously [5], the more stable mixture exhibits a narrower velocity distribution as compared to the less stable mixture. Also plotted in Fig. 3b is the 1D simulation without ozone, which shows a substantially narrower distribution of velocities as compared to the 2D simulation with the identical mixture.

To clearly demonstrate the role of confinement on velocity statistics, the probability density normalized by the probability density for the 2D wide simulation of an identical mixture $P_i(u)/P_{2D\text{ wide}}(u)$, is plotted in Fig. 3c. More specifically, the 3D square simulation is normalized by the 2D wide simulation, while the 1D simulation is normalized by the 2D wide simulation without ozone. Fig. 3c shows that the 1D simulation has less probable low and high frontal velocities compared to the 2D simulation, which has less probable low and high frontal velocities than the 3D simulation. Adding dimensionality leads to longer tailed distributions, most notable below $u/u_{CJ} = 0.8$ and above $u/u_{CJ} = 1.5$.

Interestingly, the velocity statistics in Fig. 3a and b suggest that changing confinement effectively changes detonation stability. This conclusion is supported by the observation that the 3D square velocity distribution appears similar to the 2D wide non-ozonated mixture. This means that removing confinement leads to a more globally unstable behavior, similar in nature to changing mixture chemistry. We note, however, that these detonations are still heavily influenced by their confinement, with only a few cells across channel height; it is likely that substantially larger geometries will behave differently [28].

3.2. Blast wave fitting

The velocity of strong blast waves from point-source energy release can be described using Taylor–von Neumann–Sedov theory:

$$u = c \left(\frac{E}{\rho_0} \right)^a \left(\frac{1}{t} \right)^b \quad (1)$$

where c is a constant of the order unity, E is the energy released in the blast, ρ_0 is the initial unshocked density, a is equal to $1/5$ for 3D

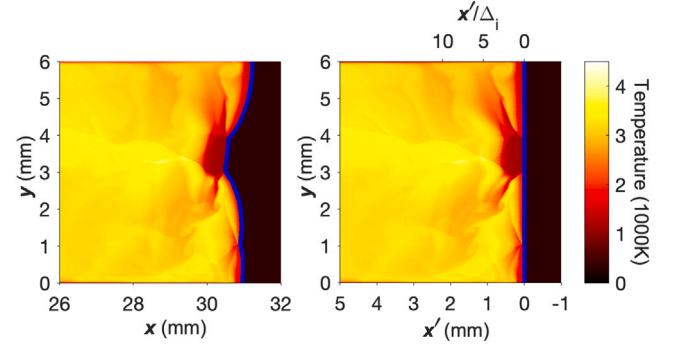


Fig. 5. Temperature pseudocolor of the same snapshot in time as in Fig. 1 of the main text of the 2D narrow case. Shown is the simulation coordinates (x and y) on the left, and the shock-flattened coordinates (x' and y') on the right.

(spherical) blasts, and $1/4$ for 2D (cylindrical) blasts, and b is equal to $3/5$ for 3D and $1/2$ for 2D [29,30]. To understand the energy release associated with kernel ignition and jetting, and to compare with theory, power law functions are fitted to the smoothed velocity decay profiles. A peak finding algorithm is used to isolate each velocity decay profile. The velocity decay is first non-dimensionalized and assumed to follow the form:

$$u/u_{CJ} = E^* \left(\frac{1}{t^* + \epsilon} \right)^b + u_{\infty}/u_{CJ} \quad (2)$$

where E^* is a non-dimensional energy release coefficient, t^* is non-dimensional time, $t^* = t/t_{\text{cell}}$, where t_{cell} is the timescale for a detonation cell, approximately $3 \mu\text{s}$, ϵ is a small coefficient to account for non-infinite initial velocity (set to 0.033), and u_{∞} is minimum velocity. Three coefficients, E^* , b , and u_{∞} are fitted by least squares regression. Of the order 10^4 velocity decay profiles are fitted for the 2D simulations, and $O(10^5)$ for the 3D simulations. The result of the decay function fitting is shown in the example profile in Fig. 2 as dashed lines, generally well-approximating the observed velocity.

Distributions of fitted parameters are estimated with the kernel density technique and shown in Fig. 4. For both 2D and 3D simulations, the fitted exponent associated with shock decay, b , is lower than predicted in theory, as shown in Fig. 4a. The theoretical model only considers energy release at a single instant, and so this slower than theoretically predicted shock decay suggests the importance of continuous heat release during the detonation. The 3D simulation shows faster shock decay as compared to the 2D simulation, as predicted in theory, and consistent with decreased dimensional confinement. Detailed examination of the fitting suggests that the bimodal behavior in the 2D b distribution is the result of multiple possible fits yielding reasonable agreement, as opposed to two physical phenomena. Given that the initial guess of the fitting algorithm is set to the theoretical value, it is likely that $b \approx 0.2$ for 2D is the most physical value. The energy release coefficients, E^* between 2D and 3D are not directly

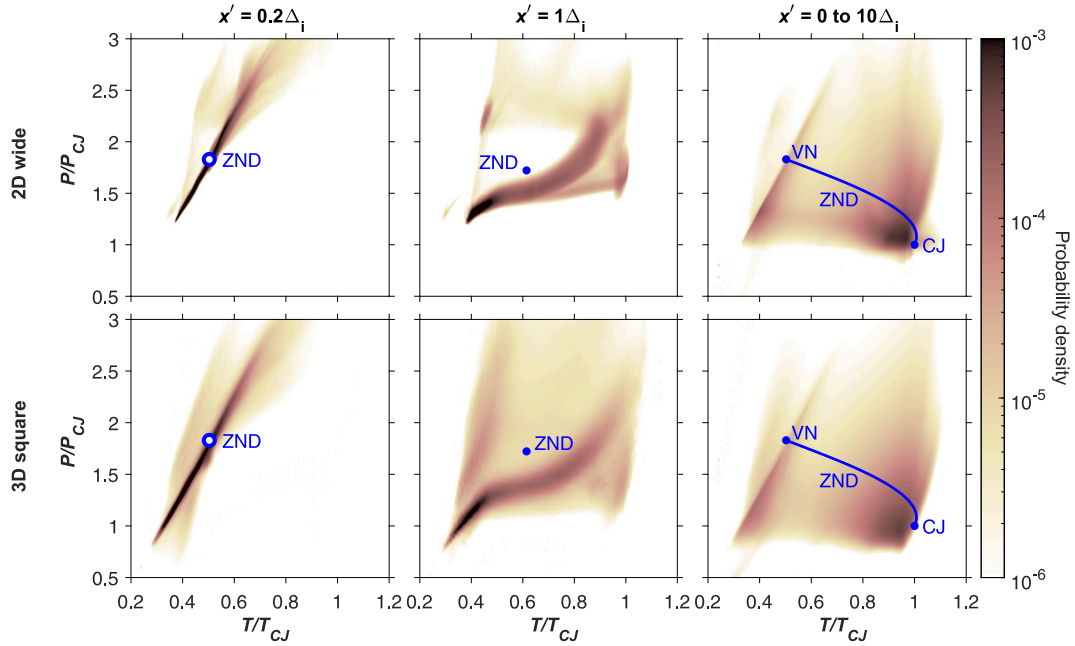


Fig. 6. 2D histograms of temperature and pressure, normalized by the CJ conditions, for the 2D wide case (top row) and 3D square case (bottom row). Selected is 2 x' locations, $0.2\Delta_i$ (first column) and $1\Delta_i$ (second column), as well as all points between 0 and $10\Delta_i$ (last column). Superimposed on the histograms are the ZND states with the von Neumann (VN) and Chapman-Jouguet (CJ) points labeled.

comparable, but we observe that the energy appears suppressed in the 2D narrow condition as compared to the 2D wide condition. E^* is very similar for the two 3D conditions. Meanwhile, the minimum velocity, u_∞ , is most often equal to 0, as in the theoretical solution.

3.3. Thermodynamic state statistics

Differences in frontal velocity due to confinement and mixture conditions can manifest in differences in thermodynamic state behind the shock and elsewhere in the domain. Thermodynamic state dictates kinetic rates, non-equilibrium timescales, real gas effects, heat transfer rates, and mechanical requirements. To study the statistical characteristics of thermodynamic state, the stream-wise direction x is transformed to distance behind the shock x' . This transformation is done for all values of y for 2D simulations and for all values of y and z for 3D simulations. This transformation is depicted in Fig. 5, showing temperature pseudocolor for the untransformed and transformed domain. In this work, temperature and pressure are used as a proxy for the full thermodynamic state, as quantified in two-dimensional histograms. Histograms are plotted both for single x' (± 2 grid points = $20 \mu\text{m}$) locations, and for all points between 0 and $10\Delta_i$. Included in these 2D histograms are $O(10^8)$ grid cells for the 3D simulations and $O(10^7)$ grid cells for the 2D simulations for the single x' location plots, and $O(10^{11})$ and $O(10^{10})$ cells for the 0 to $10\Delta_i$ histograms for the 3D and 2D simulations, respectively. We note that for the 0 to $10\Delta_i$ histograms, a more physically relevant statistical subset is all points within the separatrix [31], but due to data storage limitations, the separatrix location is not available for all simulations.

Fig. 6 shows 2D histograms for two x' locations ($0.2\Delta_i$ and $1\Delta_i$, where Δ_i is the ZND induction length, as defined in [18]) and all points between 0 and $10\Delta_i$ for the 2D wide and 3D square cases. Dictated by the frontal velocity, the thermodynamic states immediately behind the shock wave ($x' = 0.2\Delta_i$, first column of Fig. 6) are concentrated within a narrow band of pressure and temperature. This is unsurprising: temperature and pressure are coupled for a shocked gas with frozen composition. The states lie on either side of the ZND solution (blue dot) due to the cyclical nature of detonation cells (overdriven to underdriven). In the 3D square case, lower and higher temperature/pressure

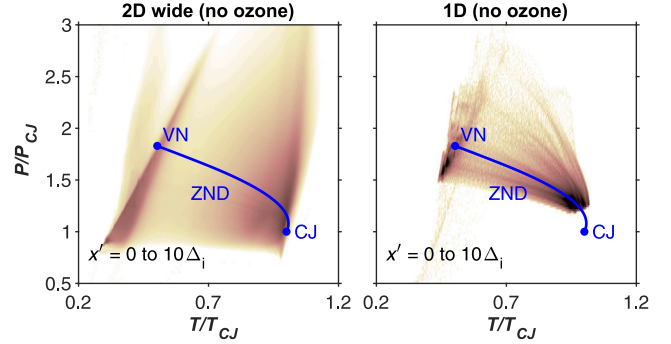


Fig. 7. 2D histograms of temperature and pressure, normalized by the CJ conditions, for the 2D wide case (left) and 1D case (right). Both cases contain no ozone additives. All x' locations between 0 and $10\Delta_i$ are considered. Superimposed on the histograms are the ZND states. Colorbar used is the same as that in Fig. 6.

conditions exist as compared to the 2D simulation, corresponding to the broader velocity distribution (see Fig. 3).

Moving further away from the shock front, corresponding, on average, to the end of the induction length ($x' = 1\Delta_i$, second column of Fig. 6), the set of thermodynamic states is broader. Although both simulations exhibit the most probable states within a similar S-shaped band, the 3D simulation has a more diffuse set of possible states. This is presumably because the velocity and blast dynamics are more variable in 3D, as described in our previous work [2]. In both cases the ZND state is improbable, likely due to negative temperature gradient forcing associated with unsteadiness, as described in [32].

The final column of Fig. 6 shows all states between 0 and $10\Delta_i$, with the ZND solution (blue line) superimposed. Notable for both simulations is that the thermodynamic states found in the multi-dimensional simulations are broader than those found in the ZND solution. The probability is highest for the burned gas region, where the bulk of the post-shock simulations lie (see Fig. 1), but is also pronounced for the post-shock region. The reaction region is somewhat less probable due to the short timescale associated with the process. As observed

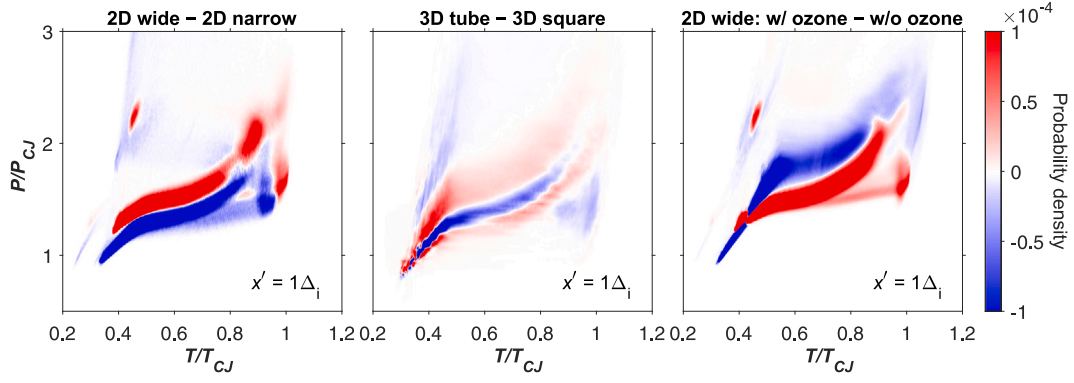


Fig. 8. Differential 2D histograms of temperature and pressure, normalized by the CJ conditions, comparing the 2D wide and 2D narrow case (first column), 3D tube and 3D square case (second column), and cases with and without ozone (third column), for $x' = 1\Delta_i$. A state red in color indicates a condition where the positive-signed simulation is more likely (2D wide, 3D tube, and 2D wide with ozone, respectively), while a state blue in color indicates a condition where the negative-signed simulation is more likely (2D narrow, 3D square, and 2D wide without ozone).

before, the 3D simulation contains a broader and more diffuse set of states as compared to the 2D simulation. Cumulative probability distributions (Fig. S1) show that extreme pressures ($P > P_{VN}$, where the P_{VN} is defined relative to a detonation propagating at the CJ velocity) constitute approximately 5% of states in both 3D simulations, but only 0.9% in the 2D wide simulation.

To determine whether this trend between 3D and 2D thermodynamic state distributions is consistent when considering 1D simulations, the thermodynamic state distribution of the 1D simulation for states between $x' = 0$ and $10\Delta_i$ is shown in Fig. 7 and compared to the identical mixture in 2D. As expected, the 1D simulation exhibits a substantially narrower distribution of thermodynamic states as compared to the 2D simulation, highlighting the importance of detonation confinement in terms of resultant thermodynamic state.

To compare the thermodynamic state distributions of similar simulations (e.g., 2D wide versus 2D narrow), differential 2D histograms are generated, which plot the differential probability, P_d :

$$P_d(T, P) = P_1(T, P) - P_2(T, P) \quad (3)$$

where $P_i(T, P)$ is the probability of a thermodynamic state as a function of temperature T and pressure P . Fig. 8 shows the differential histograms for 2D wide minus 2D narrow case, 3D tube minus 3D square, and 2D wide with ozone minus 2D wide without ozone. All cases are plotted for $x' = 1\Delta_i$. A state red in color indicates a condition where the positive-signed simulation is more likely (2D wide, 3D tube, and 2D wide with ozone, respectively), while a state blue in color indicates a condition where the negative-signed simulation is more likely (2D narrow, 3D square, and 2D wide without ozone). While qualitatively the 2D wide case looks nearly identical to the 2D narrow case, the narrow case has clearly suppressed temperatures and pressures, highlighting the role of boundary losses. These observations are consistent with both heat and momentum loss and curvature based 1D steady modeling [33,34]. The 3D tube case has a broader set of thermodynamic states as compared to the 3D square simulation. This confirms the previous observation that the 3D tube exhibits a more chaotic structure as compared to the 3D square case [2]. Finally, the 2D wide without ozone case has a different shape as compared to that with ozone. Again, the 2D wide case without ozone appears similar to the 3D square case with ozone.

Taken together, we observe that confinement plays a strong role on the overall thermodynamic state distributions exhibited by detonations. Increasing boundary losses leads to reduced velocities, temperatures, and pressures. Meanwhile, increasing dimensionality (i.e., from 1D to 2D to 3D) leads to substantially broader sets of frontal velocities and thermodynamic states, highlighting the importance of considering full 3D behavior in detonation application and science. Within the results presented here, it appears that confinement and chemistry can act

similarly, where decreasing confinement (e.g., from 2D to 3D) can effectively destabilize a detonation wave, as quantified by velocity and thermodynamic state statistics.

3.4. Grid convergence of statistics

As described in Section 2, the numerical grid convergence behavior of these simulations was extensively considered in past work [2]. Nonetheless, the statistical analysis from this work was repeated for two other grid resolutions (20 μm and 40 μm) for both the 2D wide and 3D square simulations. Fig. S2 in the Supplementary material shows the thermodynamic state distributions across various grid resolutions, and shows that while there are minor variations, the overall conclusions remain unchanged when changing grid resolutions. This analysis, as well as the previous calculations, gives confidence that the results presented here are not sensitive to numerical grid resolution.

4. Conclusions

In this work, we performed detailed statistical analysis of frontal velocity behavior and thermodynamic states for 1D, 2D, and 3D simulations, in a variety of confinements and mixture conditions. From this analysis, we see that detonation confining dimensionality and geometry effect both frontal velocity behavior and thermodynamic state statistics. More specifically, we make the following conclusions:

1. 3D simulations have a broader set of frontal velocity statistics and thermodynamic states as compared to 2D simulations. This includes higher velocities, temperatures, and pressures, existing both at the shock front and throughout the domain. Furthermore, 2D simulations have a broader set of velocities and thermodynamic state statistics than 1D simulations, which have a broader set of statistics as compared to 1D steady (ZND) simulation.
2. Decreasing detonation chemical stability (by increasing reduced activation energy ϵ_r) leads to broader velocity and thermodynamic state distributions, similar in nature to increasing dimensionality (e.g., 2D to 3D). This suggests an analog between mixture chemistry and detonation confinement.
3. Increasing relative boundary losses leads to lower temperatures, pressures, and velocities, consistent with 1D modeling.
4. The differences between the qualitative structural differences in 3D square channel and 3D round tube propagation leads to statistical differences in velocity and thermodynamic state. The 3D tube has a broader set of thermodynamic states as compared to the 3D square channel.

These conclusions do not appear sensitive to numerical grid resolution.

Novelty and Significance Statement

This research presents, to our knowledge, the first detailed statistical analysis of both frontal velocity and thermodynamic states for 3D detonation simulations. The simulations considered here constitute the state of the art—high resolution, detailed chemistry, and in large enough domains to contain several detonation cells. This work also connects, for the first time to our knowledge, the statistical analysis across multiple dimensions, confining geometries, and mixture conditions, using identical grid resolution, and where possible, mixture conditions, enabling the clear scientific conclusions presented here. This is significant because thermodynamic state (which is strongly affected by frontal velocity) constrains critical modeling physics (e.g., kinetics, real gas effects, non-equilibrium thermodynamics), and also application-related design requirements (structural, heat transfer, etc.). The conclusions from this work will inform the applicability of often-used 2D simulation and experiment for 3D models and application.

CRediT authorship contribution statement

Reza Paknahad: Performed data analysis, Developed figures, Writing – original draft. **Jonathan T. Lipkowitz:** Performed simulations, Writing – review & editing. **Noah Gaffran:** Performed data analysis, Developed figures. **Irenaeus Wlokas:** Designed simulations, Writing – review & editing. **Andreas M. Kempf:** Designed simulations, Writing – review & editing. **Jackson Crane:** Conceptualized and planned the study, Performed data analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material is included with this manuscript, describing the shock-normal velocity calculation, and presenting figures for cumulative probability distributions for temperature and pressure, and the invariance of the statistical conclusions with increasing grid resolution.

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2024.105388>.

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